

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 14

W Lecture 13:

- **Stability:** Von Neumann Ex.: 1st order linear convection/wave eqn., F-B scheme

- **Hyperbolic PDEs and Stability**

$$C = \frac{c \Delta t}{\Delta x} < 1$$

- 2nd order wave equation and waves on a string

- Characteristic finite-difference solution
 - Stability of C – C (CDS in time/space, explicit): $C = \frac{c \Delta t}{\Delta x} < 1$
 - Example: Effective numerical wave numbers and dispersion

- CFL condition:

- “Numerical domain of dependence” must include “Mathematical domain of dependence”
 - Examples: 1st order linear convection/wave eqn., 2nd order wave eqn.
 - Other FD schemes (C 2nd – C 4th)

- Von Neumann: 1st order linear convection/wave eqn., F- C: unstable

- Stability summary: Tables of schemes for 1st order linear convection/wave eqn.

- **Elliptic PDEs**

- FD schemes for 2D problems (Laplace, Poisson and Helmholtz eqns.)

- Direct 2nd order and Iterative (Jacobi, Gauss-Seidel)

- Boundary conditions

TODAY (Lecture 14): FINITE DIFFERENCES, Cont'd

- **Elliptic PDEs, Continued**

- Examples, Higher order finite differences
- Irregular boundaries: Dirichlet and Von Neumann BCs
- Internal boundaries

- **Parabolic PDEs and Stability**

- Explicit schemes (1D-space)
 - Von Neumann
- Implicit schemes (1D-space): simple and Crank-Nicholson
 - Von Neumann
- Examples
- Extensions to 2D and 3D
 - Explicit and Implicit schemes
 - Alternating-Direction Implicit (ADI) schemes

TODAY (Lecture 14, Cont'd): FINITE VOLUME METHODS

- Integral forms of the conservation laws
- Introduction to FV Methods
- Approximations needed and basic elements of a FV scheme
 - FV grids
 - Approximation of surface integrals (leading to symbolic formulas)
 - Approximation of volume integrals (leading to symbolic formulas)
- Summary: Steps to step-up FV scheme
- Examples: One Dimensional examples
 - Generic equations
 - Linear Convection (Sommerfeld eqn.): convective fluxes
 - 2nd order in space, 4th order in space, links to CDS
 - Unsteady Diffusion equation: diffusive fluxes
 - Two approaches for 2nd order in space, links to CDS

References and Reading Assignments

- Lapidus and Pinder, 1982: Numerical solutions of PDEs in Science and Engineering. Section 4.5 on “Stability”.
- Chapter 3 on “Finite Difference Methods” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 3 on “Finite Difference Approximations” of “H. Lomax, T. H. Pulliam, D.W. Zingg, *Fundamentals of Computational Fluid Dynamics (Scientific Computation)*. Springer, 2003”
- Chapter 29 and 30 on “Finite Difference: Elliptic and Parabolic equations” of “Chapra and Canale, Numerical Methods for Engineers, 2014/2010/2006.”

Elliptic PDEs

Iterative Schemes: Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Finite Difference Scheme

$$u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k - 4u_{i,j}^{k+1} = 0$$

Liebman Iterative Scheme (Jacobi/Gauss-Seidel)

$$u_{i,j}^{k+1} = u_{i,j}^k + r_{i,j}^k$$

$$r_{i,j}^k = \frac{u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k - 4u_{i,j}^k}{4}$$

$$u_{i,j}^{k+1} = \frac{u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k}{4}$$

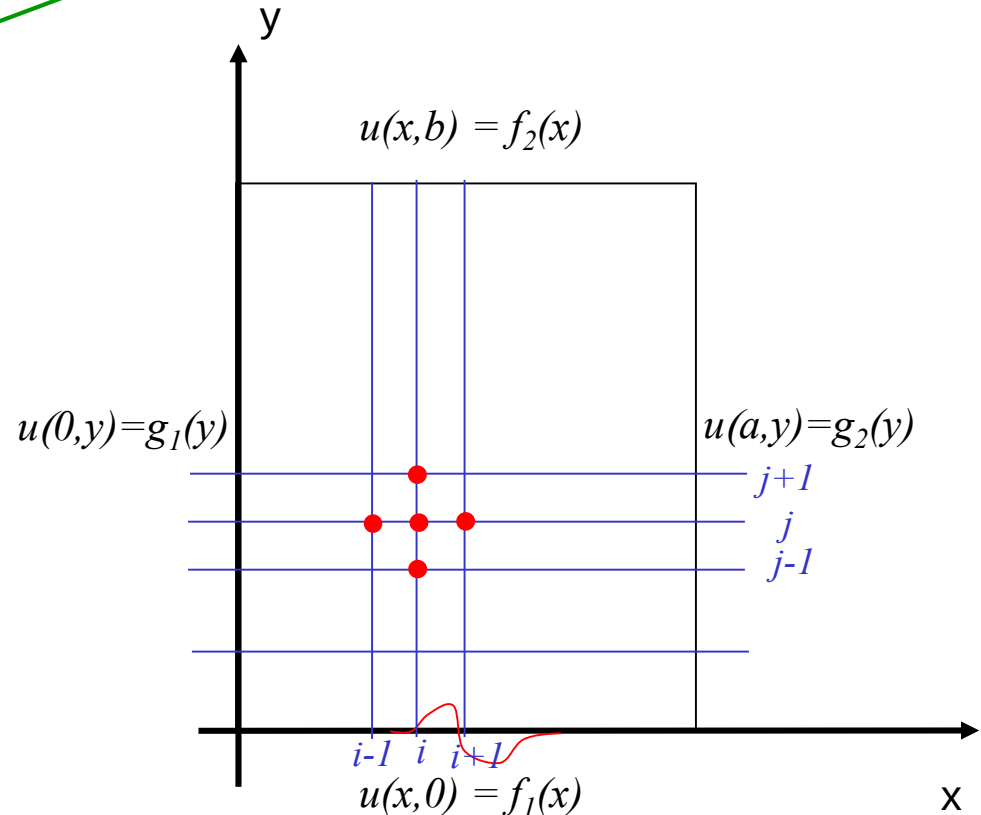
equivalent

SOR Iterative Scheme, Jacobi

$$\begin{aligned} u_{i,j}^{k+1} &= u_{i,j}^k + \omega r_{i,j}^k \\ &= u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k - 4u_{i,j}^k}{4} \\ &= (1-\omega) u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k}{4} \end{aligned}$$

Optimal SOR (Equidistant Sampling h)

$$\omega = \frac{4}{2 + \sqrt{4 - \left[\cos\left(\frac{\pi}{n-1}\right) + \cos\left(\frac{\pi}{m-1}\right) \right]^2}}$$



Elliptic PDE: Poisson Equation

$$\nabla^2 u = g(x, y)$$

$$g_{i,j} = g(x_i, y_j)$$

SOR Iterative Scheme, with Jacobi

$$\begin{aligned} u_{i,j}^{k+1} &= u_{i,j}^k + \omega r_{i,j}^k \\ &= u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k - 4u_{i,j}^k - h^2 g_{i,j}}{4} \\ &= (1 - \omega) u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^k + u_{i,j+1}^k + u_{i,j-1}^k - h^2 g_{i,j}}{4} \end{aligned}$$

Elliptic PDE: Poisson Equation

$$\nabla^2 u = g(x, y)$$

$$g_{i,j} = g(x_i, y_j)$$

SOR Iterative Scheme, with Gauss-Seidel

$$\begin{aligned} u_{i,j}^{k+1} &= u_{i,j}^k + \omega r_{i,j}^k \\ &= u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^{k+1} + u_{i,j+1}^k + u_{i,j-1}^{k+1} - 4u_{i,j}^k - h^2 g_{i,j}}{4} \\ &= (1 - \omega) u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^{k+1} + u_{i,j+1}^k + u_{i,j-1}^{k+1} - h^2 g_{i,j}}{4} \end{aligned}$$

Laplace Equation

Steady Heat diffusion (with source: Poisson eqn)

duct.m

```
Lx=1;
Ly=1;
N=10;
h=Lx/N;
M=floor(Ly/Lx*N);
niter=20;
eps=1e-6;

x=[0:h:Lx]';
y=[0:h:Ly];
f1x='4*x-4*x.^2';
% f1x='0'
f2x='0';
g1x='0';
g2x='0';
gxy='0';
f1=inline(f1x,'x');
f2=inline(f2x,'x');
g1=inline(g1x,'y');
g2=inline(g2x,'y');
gf=inline(gxy,'x','y');

n=length(x);
m=length(y);
u=zeros(n,m);
u(2:n-1,1)=f1(x(2:n-1));
u(2:n-1,m)=f2(x(2:n-1));
u(1,1:m)=g1(y);
u(n,1:m)=g2(y);
for i=1:n
    for j=1:m
        g(i,j) = gf(x(i),y(j));
    end
end
```

```
u_0=mean(u(1,:))+mean(u(n,:))+mean(u(:,1))+mean(u(:,m));
u(2:n-1,2:m-1)=u_0*ones(n-2,m-2);
omega=4/(2+sqrt(4-(cos(pi/(n-1))+cos(pi/(m-1))))^2);
for k=1:niter
    u_old=u;
    for i=2:n-1
        for j=2:m-1
            u(i,j)=(1-omega)*u(i,j)
            +omega*(u(i-1,j)+u(i+1,j)+u(i,j-1)+u(i,j+1)-h^2*g(i,j))/4;
        end
    end
    r=abs(u-u_old)/max(max(abs(u)));
    k,r
    if (max(max(r))<eps)
        break;
    end
end
figure(3)
surf(y,x,u);
shading interp;
a=ylabel('x');
set(a,'FontSize',14);
a=xlabel('y');
set(a,'FontSize',14);
a=title(['Poisson Equation - g = ' gxy]);
set(a,'FontSize',16);
```

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = g(x, y)$$

$$\text{BCs: } u(x, 0, t) = f(x) = 4x - 4x^2$$

Three other BCs are null

Helmholtz Equation

$$\nabla^2 u + f(x, y)u = g(x, y)$$

$$f_{i,j} = f(x_i, y_j)$$

$$g_{i,j} = g(x_i, y_j)$$

SOR Iterative Scheme, with Gauss-Seidel

$$\begin{aligned} u_{i,j}^{k+1} &= u_{i,j}^k + \omega r_{i,j}^k \\ &= u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^{k+1} + u_{i,j+1}^k + u_{i,j-1}^{k+1} - (4 - h^2 f_{i,j})u_{i,j}^k - h^2 g_{i,j}}{(4 - h^2 f_{i,j})} \end{aligned}$$

$$= (1 - \omega) u_{i,j}^k + \omega \frac{u_{i+1,j}^k + u_{i-1,j}^{k+1} + u_{i,j+1}^k + u_{i,j-1}^{k+1} - h^2 g_{i,j}}{(4 - h^2 f_{i,j})}$$

Elliptic PDE's

Higher Order Finite Differences

CD, 4th order (see tables in eqs. sheet)

$$\left(\frac{\partial^2 u}{\partial x^2} \right)_{\text{CD, 4th order}} = \frac{-u_{i+2,j}^k + 16u_{i+1,j}^k + 30u_{i,j}^k + 16u_{i-1,j}^k - u_{i-2,j}^k}{12h^2}$$

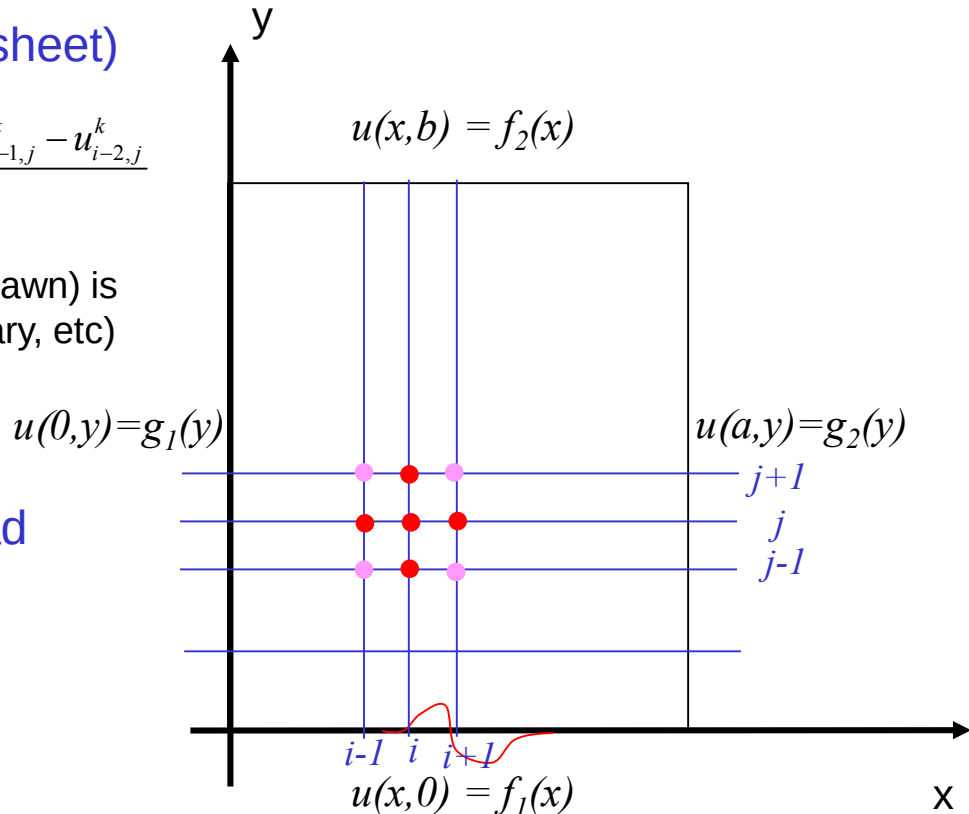
The resulting 9 point “cross” stencil (not drawn) is more challenging computationally (boundary, etc) than CD 2nd order.

Use more compact scheme instead

Square stencil (see figure):

- Use Taylor series, then cancel the terms so as to get a 4th order scheme
- Leads to:

$$\begin{aligned} \nabla^2 u_{i,j} = & \frac{1}{6h^2} [u_{i+1,j-1} + u_{i-1,j-1} + u_{i+1,j+1} + u_{i-1,j+1} \\ & + 4u_{i+1,j} + 4u_{i-1,j} + 4u_{i,j+1} + 4u_{i,j-1} - 20u_{i,j}] + O(h^4) \end{aligned}$$



Elliptic PDEs: Irregular Boundaries

Many elliptic problems don't have simple boundaries/geometries

- One way to handle them is through “irregular” discrete boundary cells (e.g. shaved cells)

1) Boundary Stencils (with Dirichlet BCs)

$$\left(\frac{\partial u}{\partial x}\right)_{i-1,i} \simeq \frac{u_{i,j} - u_{i-1,j}}{\alpha_1 \Delta x}$$

$$\left(\frac{\partial u}{\partial x}\right)_{i,i+1} \simeq \frac{u_{i+1,j} - u_{i,j}}{\alpha_2 \Delta x}$$

1st derivatives evaluated at center of edges, hence dx is sum of half edge lengths on each side

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right)$$

$$= \frac{\left(\frac{\partial u}{\partial x}\right)_{i,i+1} - \left(\frac{\partial u}{\partial x}\right)_{i-1,i}}{\frac{\alpha_1 \Delta x + \alpha_2 \Delta x}{2}}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{2}{\Delta x^2} \left[\frac{u_{i-1,j} - u_{i,j}}{\alpha_1(\alpha_1 + \alpha_2)} + \frac{u_{i+1,j} - u_{i,j}}{\alpha_2(\alpha_1 + \alpha_2)} \right]$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{2}{\Delta y^2} \left[\frac{u_{i,j-1} - u_{i,j}}{\beta_1(\beta_1 + \beta_2)} + \frac{u_{i,j+1} - u_{i,j}}{\beta_2(\beta_1 + \beta_2)} \right]$$

Can be used directly with Dirichlet BCs

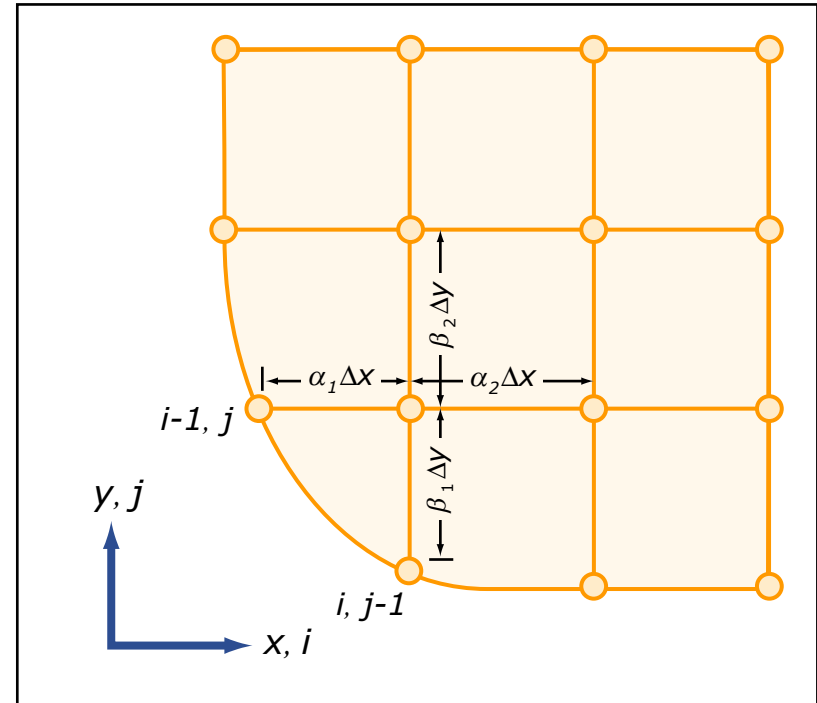


Image of a grid for a heated plate that has an irregularly shaped boundary. Image by MIT OpenCourseWare.

- Leads to direct and iterative elliptic solvers as before, but with updated coefficients for the boundary stencils
- Other options possible: curved boundary elements

Elliptic PDEs: Irregular Boundaries

2) Neumann Boundary Conditions (e.g. normal derivative given)

$$\frac{\partial u}{\partial \eta} = \frac{u_1 - u_7}{L_{17}}$$

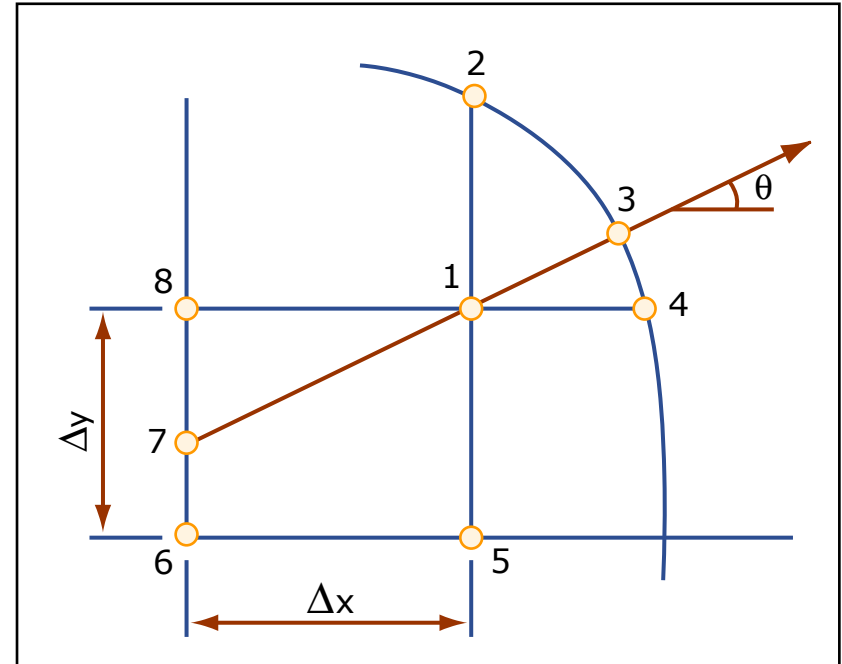
$$L_{78} = \Delta x \tan \theta$$

$$L_{17} = \Delta x / \cos \theta$$

Linear interpolation at 7

$$u_7 = u_8 + (u_6 - u_8) \frac{\Delta x \tan \theta}{\Delta y}$$

$$u_1 = \left(\frac{\Delta x}{\cos \theta} \right) \frac{\partial u}{\partial \eta} + u_6 \frac{\Delta x \tan \theta}{\Delta y} + u_8 \left(1 - \frac{\Delta x \tan \theta}{\Delta y} \right)$$



Curved boundary in which the normal gradient is specified.
Image by MIT OpenCourseWare.

- This is an approach given in Chapra & Canale
- One may instead estimate u_3 from neighbor nodes, then take the derivative along 1-3

Elliptic PDEs

Internal (Fixed) Boundaries

Velocity and Stress Continuity (heat flux or viscous stress)

$$u^+ = u^-$$

$$\mu^+ \frac{\partial u^+}{\partial y} = \mu^- \frac{\partial u^-}{\partial y}$$

Derivative Finite Differences (1st order)

$$\mu^+ \frac{\partial u^+}{\partial y} = \mu^+ \frac{u_{i,j+1} - u_{i,j}}{h}$$

$$\mu^- \frac{\partial u^-}{\partial y} = \mu^- \frac{u_{i,j} - u_{i,j-1}}{h}$$

Finite Difference Equation at bnd.

$$(\mu^- + \mu^+)u_{i,j} = \mu^+ u_{i,j+1} + \mu^- u_{i,j-1}$$

SOR Finite Difference Scheme at bnd.

$$u_{i,j}^{k+1} = (1 - \omega)u_{i,j}^k + \omega \frac{\mu^+ u_{i,j+1}^k + \mu^- u_{i,j-1}^k}{\mu^- + \mu^+}$$

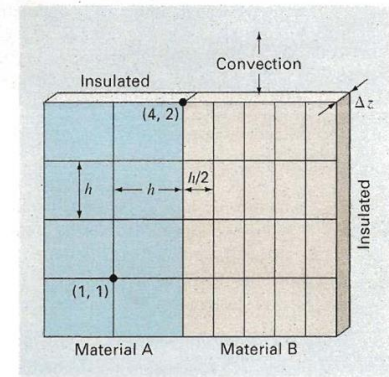
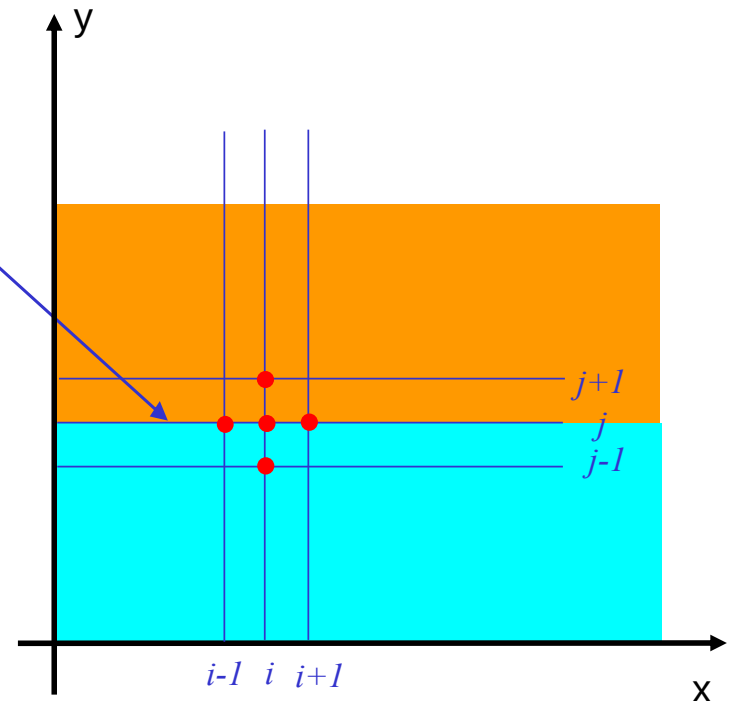


FIGURE 29.13

A heated plate with unequal grid spacing, two materials, and mixed boundary conditions.

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Source: Chapra, S. and R. Canale. *Numerical Methods for Engineers*. McGraw-Hill, 2005.



Elliptic PDEs

Internal (Fixed) Boundaries – Higher Order

Velocity and Stress Continuity

$$u^+ = u^-$$

$$\mu^+ \frac{\partial u^+}{\partial y} = \mu^- \frac{\partial u^-}{\partial y}$$

Taylor Series, inserting the PDE

$$u_{i,j-1} \simeq u_{i,j} - hu_y(x_i, y_j) + \frac{h^2}{2} u_{yy}(x_i, y_j)$$

$$= u_{i,j} - hu_y(x_i, y_j) + \frac{h^2}{2} (g_{i,j}^- - u_{xx}(x_i, y_j))$$

$$u_{i,j+1} \simeq u_{i,j} + hu_y(x_i, y_j) + \frac{h^2}{2} u_{yy}(x_i, y_j)$$

$$= u_{i,j} + hu_y(x_i, y_j) + \frac{h^2}{2} (g_{i,j}^+ - u_{xx}(x_i, y_j))$$

Derivative Finite Differences (2nd order)

$$\mu^+ \frac{\partial u^+}{\partial y} = \mu^+ \left[\frac{u_{i,j+1} - u_{i,j}}{h} + \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{2h} - \frac{h}{2} g_{i,j}^+ \right]$$

$$\mu^- \frac{\partial u^-}{\partial y} = \mu^- \left[\frac{u_{i,j} - u_{i,j-1}}{h} - \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{2h} + \frac{h}{2} g_{i,j}^- \right]$$

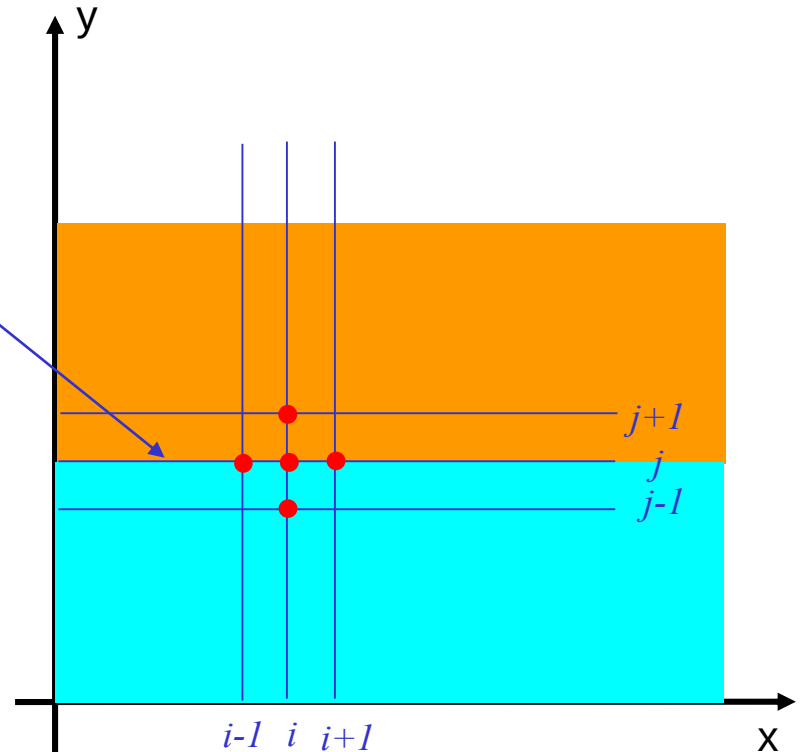
Finite Difference Equation at bnd.

$$\left[\frac{2(\mu^+ u_{i,j+1} + \mu^- u_{i,j-1}) / (\mu^- + \mu^+) + u_{i+1,j} + u_{i-1,j} - 4u_{i,j} - h^2 \bar{g}_{i,j}}{4} \right] = 0$$

SOR Finite Difference Scheme at bnd.

$$u_i^{k+1} = (1-\omega)u_i^k + \omega \left[\frac{2(\mu^+ u_{i,j+1}^k + \mu^- u_{i,j-1}^k) / (\mu^- + \mu^+) + u_{i+1,j}^k + u_{i-1,j}^k - h^2 \bar{g}_{i,j}}{4} \right]$$

$$\mu^\pm \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = f(x, y), \quad \frac{f(x, y)}{\mu^\pm} = g^\pm(x, y)$$



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Source: Chapra, S. and R. Canale. *Numerical Methods for Engineers*. McGraw-Hill, 2005.

$$\bar{g}_{i,j} = \frac{\mu^- g_{i,j}^- + \mu^+ g_{i,j}^+}{2} = \bar{f}_{i,j}$$

Partial Differential Equations

Parabolic PDE: $B^2 - 4 A C = 0$

Examples

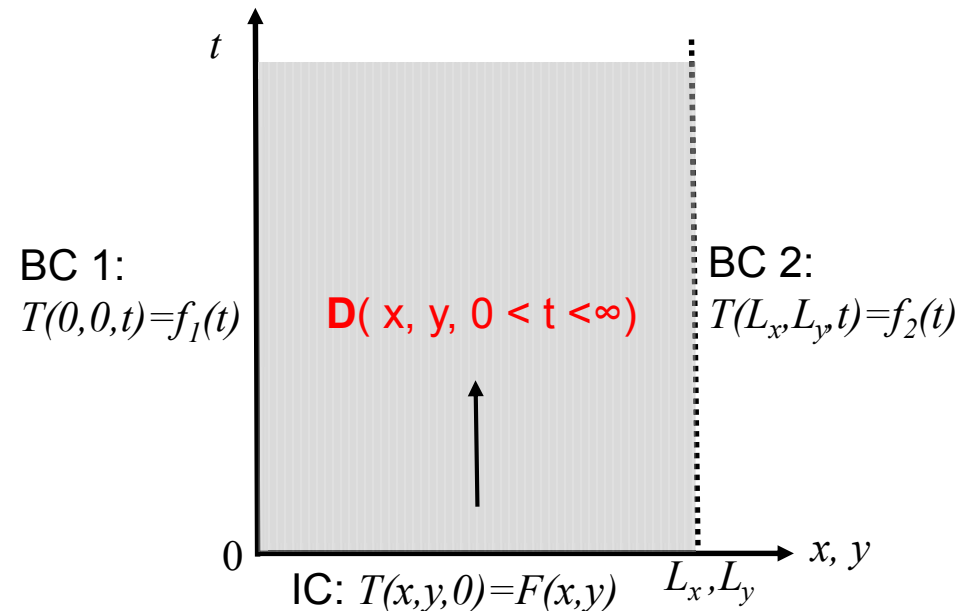
$$\frac{\partial T}{\partial t} = \frac{\kappa}{\rho c} \nabla^2 T + f, \quad (\alpha = \frac{\kappa}{\rho c})$$

Heat conduction equation,
forced or not (dominant in 1D)

$$\frac{\partial \mathbf{u}}{\partial t} = \nu \nabla^2 \mathbf{u} + \mathbf{g}$$

Unsteady, diffusive, small amplitude flows
or perturbations (e.g. Stokes Flow)

- Usually smooth solutions (“diffusion effect” present)
- “Propagation” problems
- Domain of dependence of solution is domain $\mathbf{D}$ (x, y , and $0 < t < \infty$):
- Finite Differences/Volumes, Finite Elements



Partial Differential Equations

Parabolic PDE: 1D Heat Conduction example

Heat Conduction Equation

$$T_t(x, t) = \alpha T_{xx}(x, t), 0 < x < L, 0 < t < \infty$$

$$\alpha = \frac{\kappa}{\rho c}$$

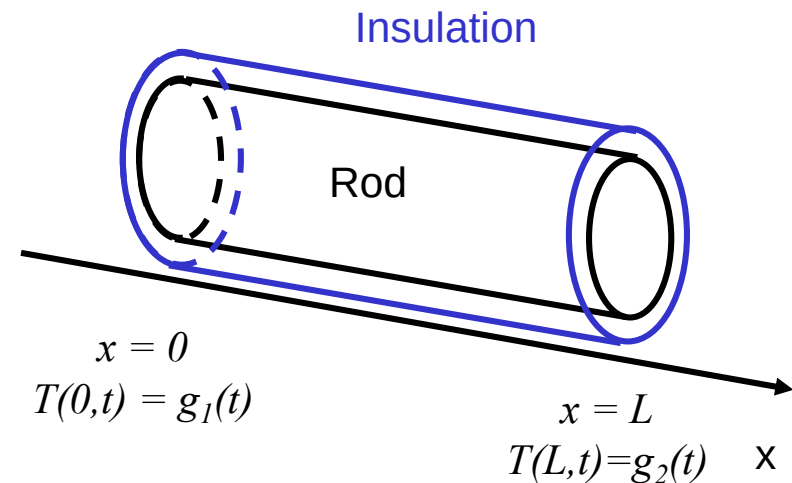
Initial Condition

$$T(x, 0) = f(x), 0 \leq x \leq L$$

Boundary Conditions

$$T(0, t) = g_1(t), 0 < t < \infty$$

$$T(L, t) = g_2(t), 0 < t < \infty$$



Parabolic PDE

1D Heat Conduction: Forward in time, centered in space, explicit

Equidistant Sampling

$$h = L/n$$

$$k = T/m$$

Discretization

$$x_i = (i-1)h, \quad i = 2, \dots, n-1$$

$$t_j = (j-1)k, \quad j = 1, \dots, m$$

Forward (Euler) Finite Difference in time

$$T_t(x, t) = \frac{T(x_i, t_{j+1}) - T(x_i, t_j)}{k} + O(k)$$

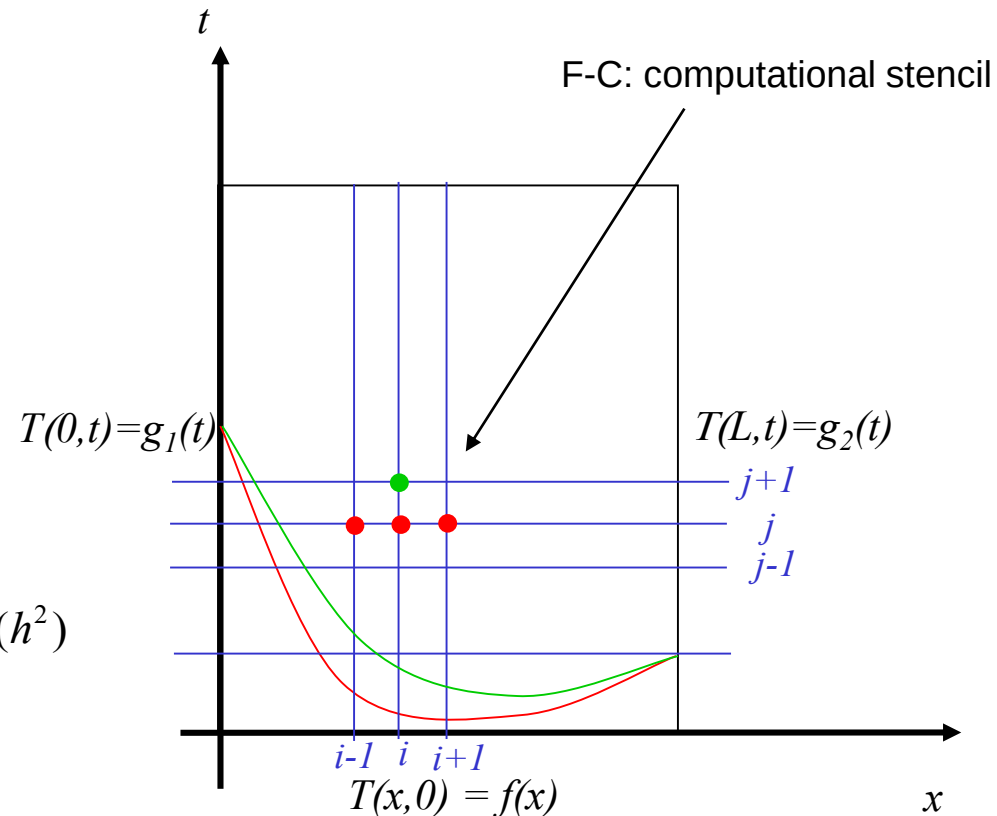
Centered Finite Difference in space

$$T_{xx}(x, t) = \frac{T(x_{i-1}, t_j) - 2T(x_i, t_j) + T(x_{i+1}, t_j))}{h^2} + O(h^2)$$

$$T_{i,j} = T(x_i, t_j)$$

Finite Difference Equation

$$\frac{T_{i,j+1} - T_{i,j}}{k} = \alpha \frac{T_{i-1,j} - 2T_{i,j} + T_{i+1,j}}{h^2}$$



Parabolic PDE

1D Heat Conduction: Forward in time, centered in space, explicit

Dimensionless diffusion coefficient

$$r = \frac{\alpha k}{h^2}$$

Explicit Finite Difference Scheme

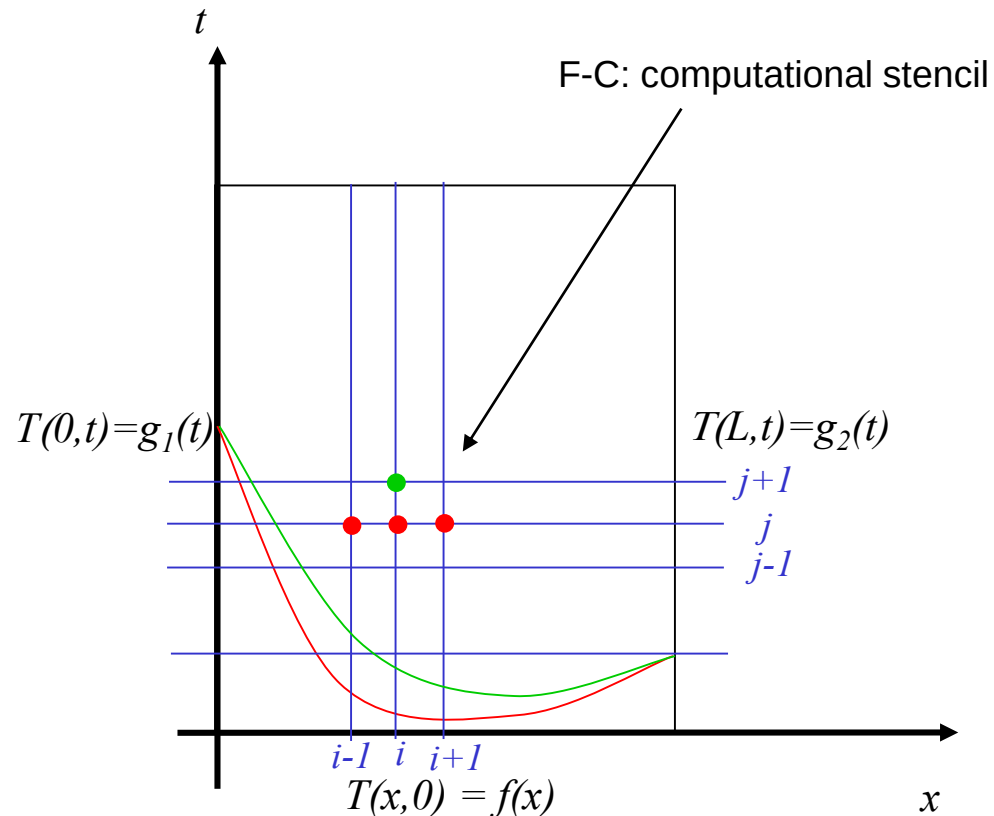
$$T_{i,j+1} = (1 - 2r)T_{i,j} + r(T_{i-1,j} + T_{i+1,j})$$

Stability Requirement

$$r \leq 0.5$$

Conditionally stable
(von Neumann)

Shown in class on blackboard



Heat Conduction Equation

Explicit Finite Differences

(1D-in-space, unsteady case;
similar to steady elliptic problem seen previously)

T denoted by u , i.e. $u_{i,j} \equiv T_{i,j}$

α denoted by c^2 , i.e. $c^2 \equiv \alpha$

heat_fw.m

```
L=1; T=0.2; c=1;
N=5; h=L/N;
M=10; k=T/M;
r=c^2*k/h^2

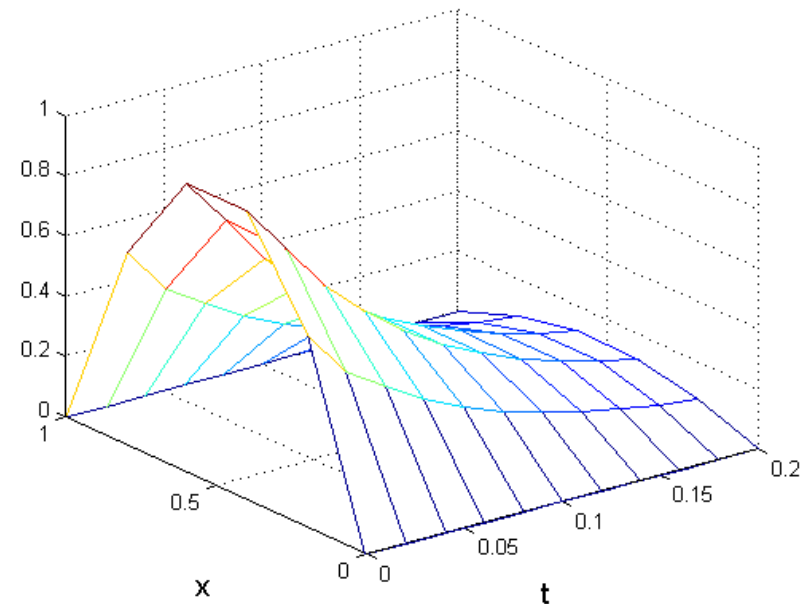
x=[0:h:L]';
t=[0:k:T];
fx='4*x-4*x.^2';
glx='0';
g2x='0';
f=inline(fx,'x');
gl=inline(glx,'t');
g2=inline(g2x,'t');
n=length(x);
m=length(t);
u=zeros(n,m);
u(2:n-1,1)=f(x(2:n-1));
u(1,1:m)=gl(t);
u(n,1:m)=g2(t);
for j=1:m-1
    for i=2:n-1
        u(i,j+1)=(1-2*r)*u(i,j) + r*(u(i+1,j)+u(i-1,j));
    end
end
figure(4)
mesh(t,x,u);
a=ylabel('x');
set(a,'FontSize',14);
a=xlabel('t');
set(a,'FontSize',14);
a=title(['Forward Euler - r =' num2str(r)]);
set(a,'FontSize',16);
```

$$u_t(x,t) = u_{xx}(x,t), \quad 0 < x < 1, \quad 0 < t < 0.2$$

ICs: $u(x,0) = f(x) = 4x - 4x^2$

BCs: $u(0,t) = g_1(t) \equiv 0$
 $u(1,t) = g_2(t) \equiv 0$

Forward Euler - r=0.5



Heat Conduction Equation

Explicit Finite Differences

```
L=1; T=0.333; c=1;
N=5; h=L/N;
M=10; k=T/M;
r=c^2*k/h^2
```

heat_fw_2.m

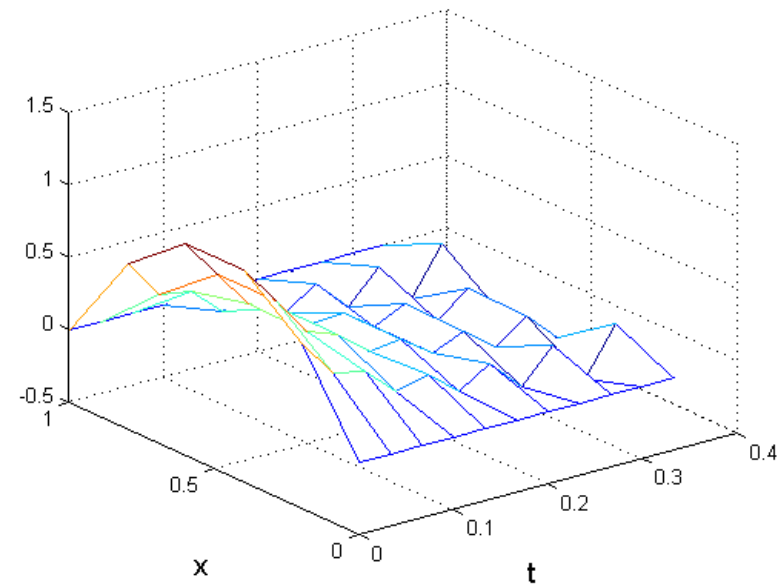
```
x=[0:h:L]';
t=[0:k:T];
fx='4*x-4*x.^2';
glx='0';
g2x='0';
f=inline(fx,'x');
gl=inline(glx,'t');
g2=inline(g2x,'t');
n=length(x);
m=length(t);
u=zeros(n,m);
u(2:n-1,1)=f(x(2:n-1));
u(1,1:m)=gl(t);
u(n,1:m)=g2(t);
for j=1:m-1
    for i=2:n-1
        u(i,j+1)=(1-2*r)*u(i,j) + r*(u(i+1,j)+u(i-1,j));
    end
end
figure(4)
mesh(t,x,u);
a=ylabel('x');
set(a,'FontSize',14);
a=xlabel('t');
set(a,'FontSize',14);
a=title(['Forward Euler - r =' num2str(r)]);
set(a,'FontSize',16);
```

$$u_t(x,t) = u_{xx}(x,t), \quad 0 < x < 1, \quad 0 < t < 0.33$$

ICs: $u(x,0) = f(x) = 4x - 4x^2$

BCs: $u(0,t) = g_1(t) \equiv 0$
 $u(1,t) = g_2(t) \equiv 0$

Forward Euler - r=0.8325

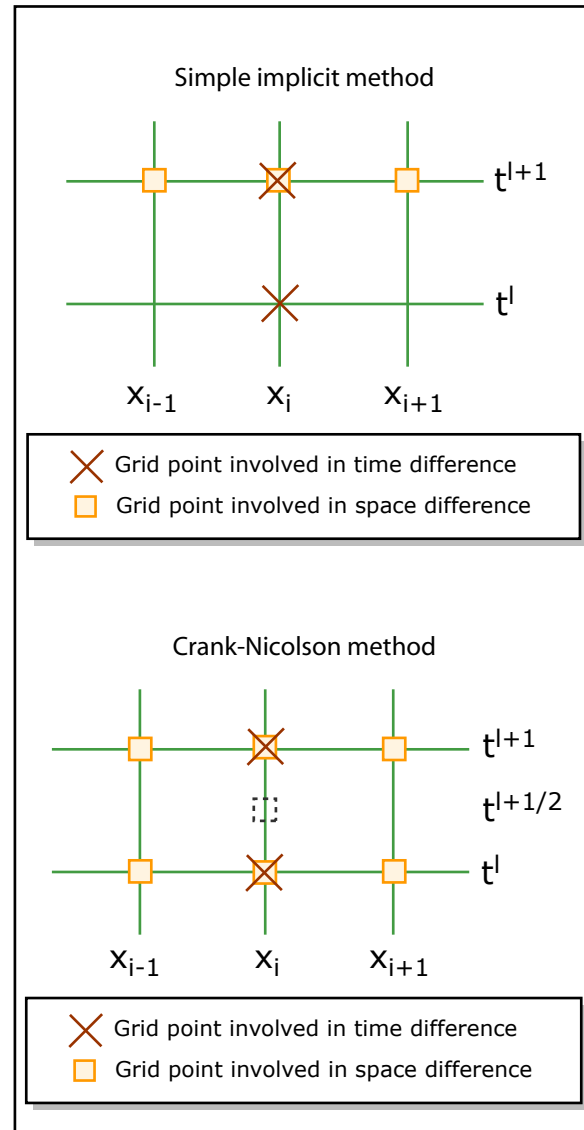


Parabolic PDE: Implicit Schemes

Leads to a system of equations to be solved at each time-step

B-C (Backward-Centered):
1st order accurate in time,
2nd order in space

Unconditionally stable



Crank-Nicolson:
2nd order accurate in time,
2nd order in space

Unconditionally stable

B-C:

- Backward in time
- Centered in space
- Evaluates RHS at time $t+1$ instead of time t (for the explicit scheme)

Time: centered FD, but evaluated at mid-point

2nd derivative in space determined at mid-point by averaging at t and $t+1$

Image by MIT OpenCourseWare. After Chapra, S., and R. Canale.
Numerical Methods for Engineers. McGraw-Hill, 2005.

Parabolic PDE: Implicit Schemes

Crank-Nicolson Scheme

Equidistant Sampling

$$h = L/n$$

$$k = T/m$$

Discretization

$$x_i = (i-1)h, \quad i = 2, \dots, n-1$$

$$t_j = (j-1)k, \quad j = 1, \dots, m$$

Mid-point Finite Difference in time

$$u_t \left(x, t + \frac{k}{2} \right) = \frac{u(x, t+k) - u(x, t)}{k} + O(k^2)$$

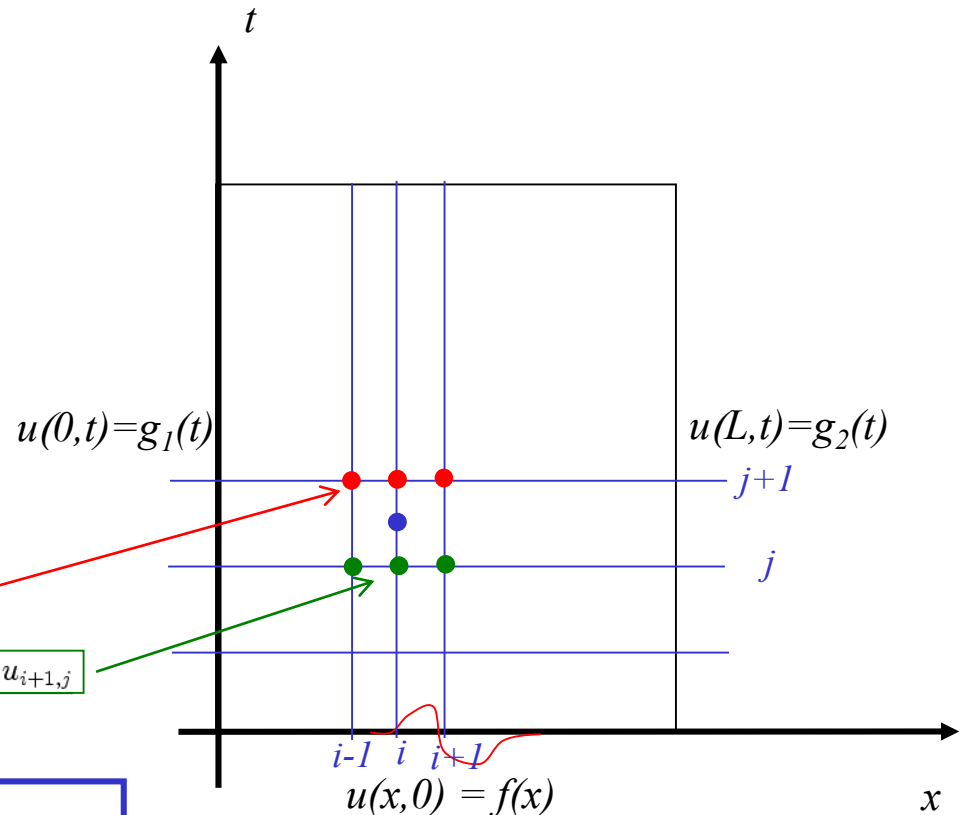
Finite Difference Equation

$$\frac{u_{i,j+1} - u_{i,j}}{k} = c^2 \left[\frac{u_{i-1,j+1} - 2u_{i,j+1} + u_{i+1,j+1}}{2h^2} + \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{2h^2} \right]$$

Crank-Nicolson Implicit Scheme

$$r = \frac{c^2 k}{h^2}$$

$$-ru_{i-1,j+1} + (2+2r)u_{i,j+1} - ru_{i+1,j+1} = (2-2r)u_{i,j} + r(u_{i-1,j} + u_{i+1,j})$$



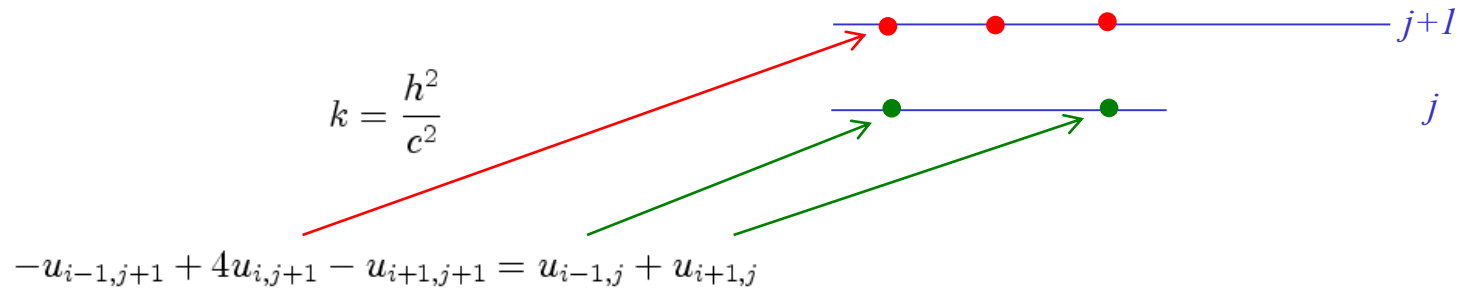
Unconditionally stable
(by Von Neumann)

Parabolic PDEs: Implicit Schemes

Crank-Nicolson – special case of $r = 1$

$$r = \frac{c^2 k}{h^2} = 1$$

$$k = \frac{h^2}{c^2}$$



$$\begin{bmatrix} 4 & -1 & & & 0 \\ -1 & 4 & -1 & & \\ & & \cdot & \cdot & \\ & -1 & 4 & -1 & \\ & & \cdot & \cdot & \\ 0 & & -1 & 4 & -1 \\ & & & -1 & 4 \end{bmatrix} \begin{Bmatrix} u_{2,j+1} \\ u_{3,j+1} \\ \cdot \\ u_{i,j+1} \\ \cdot \\ u_{n-2,j+1} \\ u_{n-1,j+1} \end{Bmatrix} = \begin{Bmatrix} g_{1,j} + u_{3,j} + g_{1,j+1} \\ u_{2,j} + u_{4,j} \\ \cdot \\ u_{i-1,j} + u_{i+1,j} \\ \cdot \\ u_{n-3,j} + u_{n-1,j} \\ u_{n-2,j} + g_{n,j} + g_{n,j+1} \end{Bmatrix}$$

Heat Flow Equation

Implicit Crank-Nicolson Scheme

```

L=1; T=0.333; c=1;
N=5; h=L/N;
M=10;
k=T/M;
r=c^2*k/h^2

                                heat_cn.m

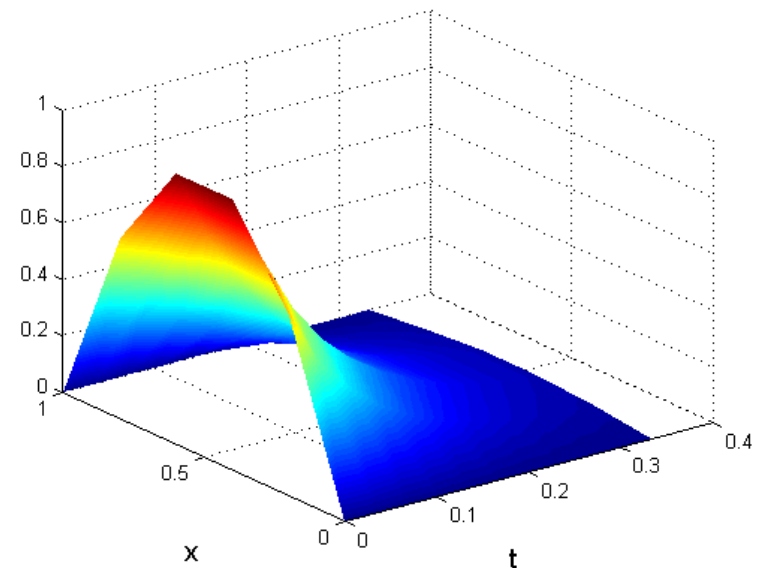
x=[0:h:L]';
t=[0:k:T];
fx='4*x-4*x.^2';
g1x='0';
g2x='0';
f=inline(fx,'x');
g1=inline(g1x,'t');
g2=inline(g2x,'t');
n=length(x); m=length(t); u=zeros(n,m);
u(2:n-1,1)=f(x(2:n-1));
u(1,1:m)=g1(t); u(n,1:m)=g2(t);
% set up Crank-Nicolson coef matrix
d=(2+2*r)*ones(n-2,1);
b=-r*ones(n-2,1);
c=b;
% LU factorization
[alf,bet]=lu_tri(d,b,c);
for j=1:m-1
    rhs=r*(u(1:n-2,j)+u(3:n,j)) + (2-2*r)*u(2:n-1,j);
    rhs(1) = rhs(1)+r*u(1,j+1);
    rhs(n-2)=rhs(n-2)+r*u(n,j+1);
% Forward substitution
    z=forw_tri(rhs,bet);
% Back substitution
    y_b=back_tri(z,alf,c);
    for i=2:n-1
        u(i,j+1)=y_b(i-1);
    end
end
    
```

$$u_t(x, t) = u_{xx}(x, t), \quad 0 < x < 1, \quad 0 < t < 0.0$$

ICs: $u(x, 0) = f(x) = 4x - 4x^2$

BCs: $u(0, t) = g_1(t) \equiv 0$
 $u(1, t) = g_2(t) \equiv 0$

Crank-Nicholson - r = 0.8325



Heat Flow Equation

Implicit Crank-Nicolson Scheme

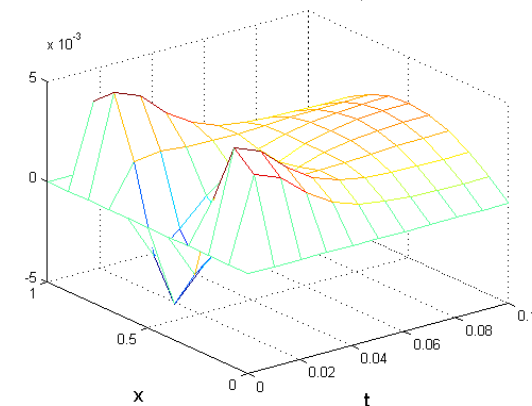
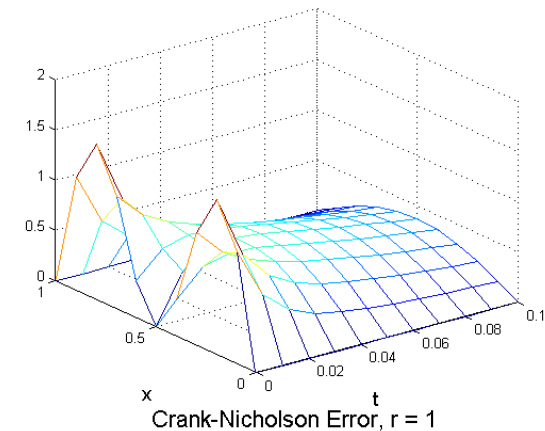
Initial Condition

$$f(x) = \sin \pi x + \sin 3\pi x$$

Analytical Solution

$$u(x, t) = e^{-\pi^2 t} \sin \pi x + e^{-9\pi^2 t} \sin 3\pi x$$

Crank-Nicolson - $r=1$



```

L=1; T=0.1; c=1;
N=10; h=L/N;
M=10;
k=T/M;
r=c^2*k/h^2

heat_cn_sin.m

x=[0:h:L]';
t=[0:k:T];
fx='sin(pi*x)+sin(3*pi*x)';
g1x='0';
g2x='0';
f=inline(fx,'x');
g1=inline(g1x,'t');
g2=inline(g2x,'t');
n=length(x); m=length(t); u=zeros(n,m);
u(2:n-1,1)=f(x(2:n-1));
u(1,1:m)=g1(t); u(n,1:m)=g2(t);
% set up Crank-Nicolson coef matrix
d=(2+2*r)*ones(n-2,1);
b=-r*ones(n-2,1);
c=b;
% LU factorization
[alf,bet]=lu_tri(d,b,c);
for j=1:m-1
    rhs=r*(u(1:n-2,j)+u(3:n,j)) + (2-2*r)*u(2:n-1,j);
    rhs(1) = rhs(1)+r*u(1,j+1);
    rhs(n-2)=rhs(n-2)+r*u(n,j+1);
% Forward substitution
    z=forw_tri(rhs,bet);
% Back substitution
    y_b=back_tri(z,alf,c);
    for i=2:n-1
        u(i,j+1)=y_b(i-1);
    end
end
end
    
```

Parabolic PDEs: Two spatial dimensions

- Example: Heat conduction equation/unsteady diffusive (e.g. negligible flow, no convection)

$$\frac{\partial T}{\partial t} = c^2 \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right), \quad (0 \leq t < \infty, 0 < x < L_x, 0 < y < L_y)$$

- Standard explicit and implicit schemes ($t = n\Delta t$, $x = i\Delta x$, $y = j\Delta y$)

- Explicit:
$$\frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = c^2 \frac{T_{i-1,j}^n - 2T_{i,j}^n + T_{i+1,j}^n}{\Delta x^2} + c^2 \frac{T_{i,j-1}^n - 2T_{i,j}^n + T_{i,j+1}^n}{\Delta y^2} \quad (O(\Delta t), O(\Delta x^2 + \Delta y^2))$$

- Stringent stability criterion:

$$\Delta t \leq \frac{1}{8} \frac{\Delta x^2 + \Delta y^2}{c^2} \quad \text{For uniform grid: } r = \frac{\Delta t c^2}{\Delta x^2} \leq \frac{1}{4}$$

- Implicit:
$$\frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = c^2 \frac{T_{i-1,j}^{n+1} - 2T_{i,j}^{n+1} + T_{i+1,j}^{n+1}}{\Delta x^2} + c^2 \frac{T_{i,j-1}^{n+1} - 2T_{i,j}^{n+1} + T_{i,j+1}^{n+1}}{\Delta y^2} \quad (O(\Delta t), O(\Delta x^2 + \Delta y^2))$$

- Crank-Nicolson Implicit (for $\Delta x = \Delta y$) $(O(\Delta t^2), O(\Delta x^2))$

$$(1 + 2r)T_{i,j}^{n+1} - (1 - 2r)T_{i,j}^n = \frac{r}{2} (T_{i-1,j}^{n+1} + T_{i+1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1}) + \frac{r}{2} (T_{i-1,j}^n + T_{i+1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n)$$

- Centered in time over Δt = Sum explicit and implicit RHSs (given above), divided by two

Parabolic PDEs: Two spatial dimensions

- Crank-Nicolson Implicit (for $\Delta x = \Delta y$):

$$(1 + 2r)T_{i,j}^{n+1} - (1 - 2r)T_{i,j}^n = \frac{r}{2}(T_{i-1,j}^{n+1} + T_{i+1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1}) + \frac{r}{2}(T_{i-1,j}^n + T_{i+1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n)$$

- Five unknowns at the (n+1) time => penta-diagonal
- Either elimination procedure or iterative scheme (Jacobi/Gauss-Seidel/SOR)
- but not always efficient

- Alternating-Direction Implicit (ADI) schemes

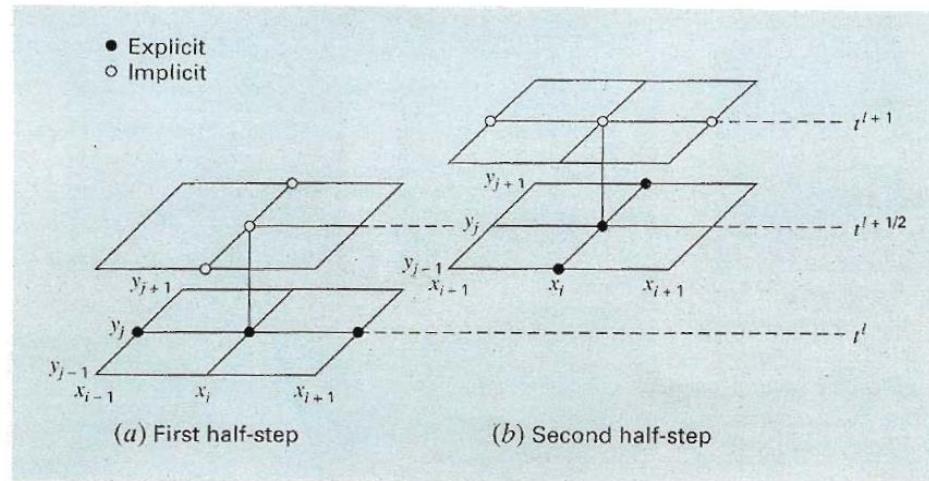
- Provides a mean for solving parabolic PDEs with tri-diagonal matrices
- In 2D: each time increment is executed in two half steps: each step is conditionally stable, but “combination of two half-steps” is unconditionally stable (similar to Crank-Nicolson behavior)
- It is one but a group of schemes called “splitting methods”
- Extended to 3D (time increment divided in 3): varied stability properties

Parabolic PDEs: Two spatial dimensions

ADI scheme (Two Half steps in time)

FIGURE 30.10

The two half-steps used in implementing the alternating-direction implicit scheme for solving parabolic equations in two spatial dimensions.



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Source: Chapra, S., and R. Canale. *Numerical Methods for Engineers*. McGraw-Hill, 2005.

- 1) From time n to $n+1/2$: Approximation of 2nd order x derivative is explicit, while the y derivative is implicit. Hence, tri-diagonal matrix to be solved:

$$\frac{T_{i,j}^{n+1/2} - T_{i,j}^n}{\Delta t / 2} = c^2 \frac{T_{i-1,j}^n - 2T_{i,j}^n + T_{i+1,j}^n}{\Delta x^2} + c^2 \frac{T_{i,j-1}^{n+1/2} - 2T_{i,j}^{n+1/2} + T_{i,j+1}^{n+1/2}}{\Delta y^2} \quad (O(\Delta x^2 + \Delta y^2))$$

- 2) From time $n+1/2$ to $n+1$: Approximation of 2nd order x derivative is implicit, while the y derivative is explicit. Another tri-diagonal matrix to be solved:

$$\frac{T_{i,j}^{n+1} - T_{i,j}^{n+1/2}}{\Delta t / 2} = c^2 \frac{T_{i-1,j}^{n+1} - 2T_{i,j}^{n+1} + T_{i+1,j}^{n+1}}{\Delta x^2} + c^2 \frac{T_{i,j-1}^{n+1/2} - 2T_{i,j}^{n+1/2} + T_{i,j+1}^{n+1/2}}{\Delta y^2} \quad (O(\Delta x^2 + \Delta y^2))$$

Parabolic PDEs: Two spatial dimensions

ADI scheme (Two Half steps in time)

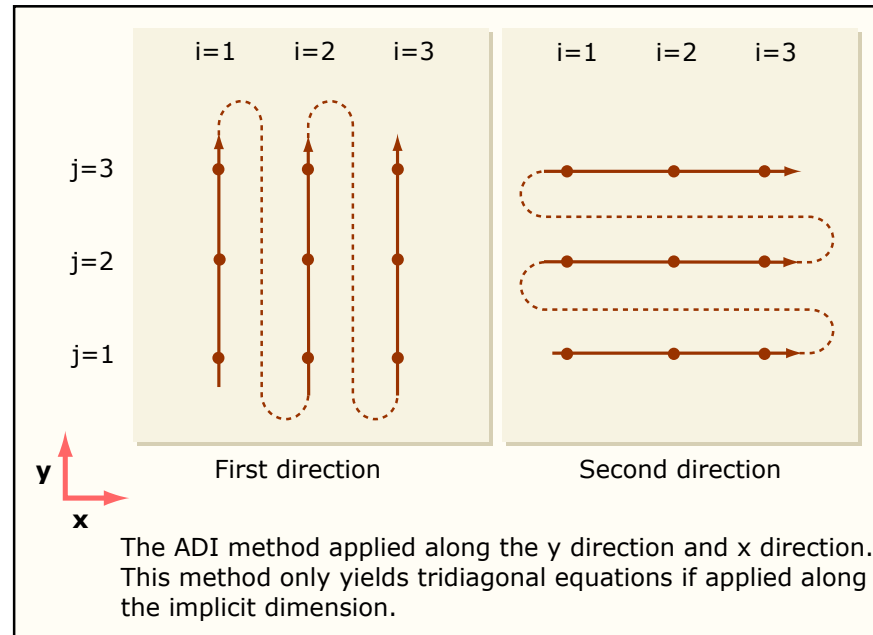


Image by MIT OpenCourseWare. After Chapra, S., and R. Canale. *Numerical Methods for Engineers*. McGraw-Hill, 2005.

For $\Delta x = \Delta y$:

1) From time n to $n+1/2$:

$$-rT_{i,j-1}^{n+1/2} + 2(1+r)T_{i,j}^{n+1/2} - rT_{i,j+1}^{n+1/2} = rT_{i-1,j}^n + 2(1-r)T_{i,j}^n + rT_{i+1,j}^n$$

2) From time $n+1/2$ to $n+1$:

$$-rT_{i-1,j}^{n+1} + 2(1+r)T_{i,j}^{n+1} - rT_{i+1,j}^{n+1} = rT_{i,j-1}^{n+1/2} + 2(1-r)T_{i,j}^{n+1/2} + rT_{i,j+1}^{n+1/2}$$

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2.29 Numerical Fluid Mechanics

Spring 2015

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